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Comprehensive Project Proposal: Brain MRI Augmentation

1. Project Selection

Selected Project: 2. Brain MRI Augmentation

2. Problem Statement

Explain the real problem you want to solve:

- **What is the issue?**
  Modern medical diagnosis increasingly relies on Deep Learning (DL) models for tasks like brain tumor segmentation and classification. However, these models require massive volumes of high-quality, diverse, and labeled MRI data. In practice, medical datasets are frequently small, highly imbalanced (e.g., fewer samples of rare tumor types), and strictly protected by privacy regulations like HIPAA, making data sharing across institutions extremely difficult.
- **Why is it important?**
  Training on small datasets leads to "overfitting," where the model learns the noise of the specific data rather than general medical features. This results in poor generalization, meaning an AI might work in one hospital but fail in another. Robust data augmentation is essential to ensure that diagnostic tools are reliable, safe, and unbiased across different patient populations.
- **Who is affected?**
  1. **Medical Researchers:** They are hindered by the lack of training samples for rare pathologies.
  2. **Radiologists:** They face higher workloads without reliable automated "second-opinion" tools.
  3. **Patients:** They are at risk of misdiagnosis if an AI tool has not seen enough diverse anatomical variations during its training phase.
- **What happens if the problem is not solved?**
  Without a solution for data scarcity, the "AI gap" in healthcare will widen. Clinical adoption of AI will remain low due to lack of trust in model reliability. Furthermore, rare brain conditions will remain under-researched simply because there isn't enough data to train predictive algorithms.

3. Model Selection

Chosen Model: GAN (Deep Convolutional Generative Adversarial Network - DCGAN)
Why?
We have selected the **DCGAN** architecture for several critical reasons:

1. **Adversarial Learning:** Unlike Variational Autoencoders (VAEs), which use a pixel-wise loss that often results in "blurry" images, GANs use a discriminator that forces the generator to produce sharp, high-frequency details essential for medical textures.
2. **Spatial Hierarchy:** DCGAN replaces pooling layers with strided convolutions (Discriminator) and fractional-strided convolutions (Generator), allowing the network to learn its own spatial downsampling and upsampling, which is superior for maintaining the anatomical geometry of the brain.
3. **Stability:** By using Batch Normalization and LeakyReLU activations, DCGAN provides a more stable training environment than traditional GANs, which is vital when working with complex grayscale medical intensities.

4. Dataset Details

- **Dataset Name:** Brain Tumor MRI Dataset (Consolidated Collection)
- **Source:** Kaggle / Figshare Public Medical Repositories
- **Number of Samples:** ~5,712 images (distributed across 4 classes)
- **Data Type:** Image (Grayscale MRI - Axial View)
- **Public or Private:** Public
- **Classes:** Glioma, Meningioma, Pituitary Tumor, and No Tumor.

5. Methodology (Step-by-Step)

1. **Collect Dataset:** Gather the raw MRI scans in JPG/PNG format. Organize them into directories corresponding to their pathological labels to facilitate supervised training logic if needed.
2. **Clean and Preprocess Data:**
    - **Resizing:** Uniformly resize all images to 128x128 pixels.
    - **Grayscale Conversion:** Ensure all images are single-channel to focus on structural density.
    - **Normalization:** Rescale pixel values from [0, 255] to [-1, 1] to match the Tanh activation function of the generator.
    - **Augmentation:** Apply random horizontal flips to increase the diversity of the "real" training set.
3. **Build Model Architecture:**
    - **Generator:** Design a network that takes a 100-dimension latent vector and passes it

through 6 transposed convolution layers with Batch Normalization and ReLU.
    - **Discriminator:** Design a CNN that takes a 128x128 image and uses strided convolutions with LeakyReLU (0.2) to output a single probability score.
4. **Train Model:**
   Implement the minimax game where the Discriminator is trained to maximize the probability of assigning the correct label, while the Generator is trained to minimize $\log(1 - D(G(z)))$. Use Adam optimizer with a learning rate of 0.0002 and $\beta_1 = 0.5$.
5. **Test Model:**
   Generate batches of images using a "fixed noise" vector at the end of every 10 epochs to track visual progress and ensure the model is not suffering from mode collapse.
6. **Evaluate Results:**
   Quantify the quality using FID (to check distribution) and SSIM (to check structural consistency). Compare generated tumor characteristics against real samples.
7. **Build Interface/Demo:**
   Create a **Gradio** web application that loads the trained .pth weights. The UI will feature a "Generate" button, a slider for the number of images, and a gallery view to display the synthetic MRIs.
8. **Final Report:**
   Synthesize all findings into a technical document, including Loss vs. Epoch graphs and a comparison of real vs. synthetic images.

6. Success Metrics

To rigorously measure the performance of our DCGAN, we will use:

- **FID (Fréchet Inception Distance):** * *Definition:* Calculates the distance between feature vectors of real and fake images.
    - *Target:* A score below 50 indicates high-quality distribution matching.
- **SSIM (Structural Similarity Index):** * *Definition:* Focuses on luminance, contrast, and structure.
    - *Target:* Scores above 0.70 indicate that the synthetic brain retains realistic anatomical structures.
- **PSNR (Peak Signal-to-Noise Ratio):** * *Definition:* Measures the ratio between the maximum possible power of a signal and the power of corrupting noise.
    - *Target:* Higher values (typically > 20dB) indicate clearer, less noisy image generation.
- **Qualitative Visual Inspection:** * A "Turing Test" for images where generated samples are compared side-by-side with real ones to ensure no obvious checkerboard artifacts or anatomical impossibilities.